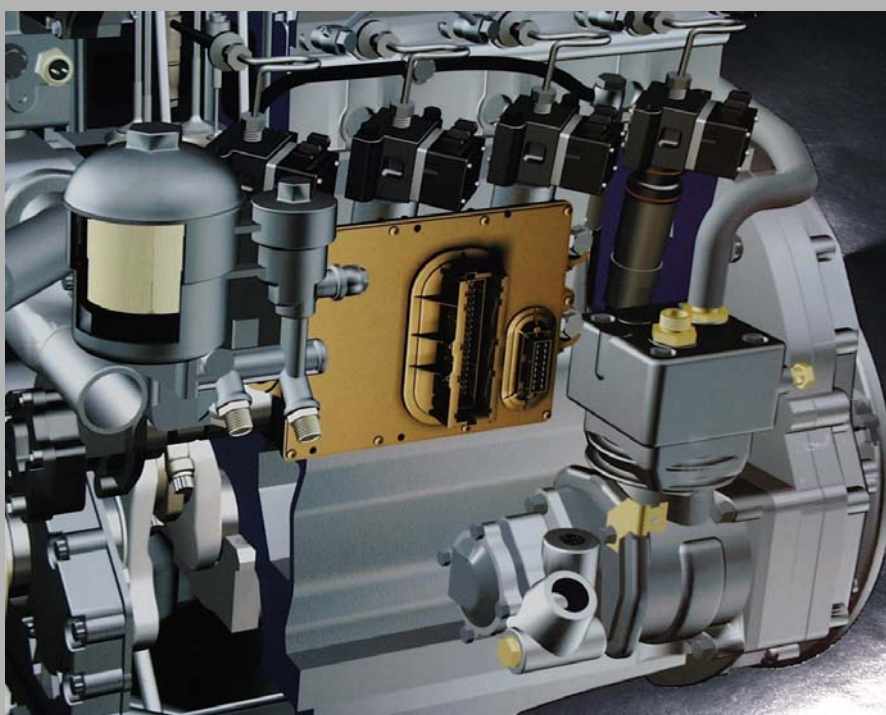




Mercedes-Benz



Operating Instructions for Engine Control Unit MR2A/MR2B

Document information

Applicability

Applicability	Control unit versions from MR2A1/MR2B1.2 to MR2A1.4/MR2B1.4
Name of document	Product documentation for MR2A/MR2B
Project	DC-MR2
Use	Generally accessible
Number of pages	175
File name	Produktdokumentation MR2.pdf

	Name:	Department:	Tel:	Date:	Signature
Compiled by		TPE/PEE	53016	14.09.07	
Checked by					
Approved by					
Responsibility of					

List of versions

Version	Date	Compiled by	Comments/description
1.0	14.09.2007	Vornberger	Preliminary version (draft)

Revision levels

Version	Date	Compiled by	Amendment

Environmental protection

Daimler's declared policy is one of integrated environmental protection that begins at the roots of the factors which affect the environment and takes all the ecological effects of its production processes and its products into account in corporate decision-making.

The objectives are for the natural resources which form the basis of our existence on this planet to be used sparingly and in a manner which takes the requirements of both nature and humanity into account.

Thank you for choosing a Mercedes-Benz product.

These Operating Instructions describe the MR2A and MR2B engine control units and are intended to help you understand the technology contained in these control units. It also provides you with important information regarding diagnosis and any troubleshooting procedures that you may need to carry out.

You can obtain assistance and advice at any time from a Mercedes-Benz Service Center.

These Operating Instructions also make reference to the Operating Instructions for the "MR-PLD engine control unit". Please keep these accessible.

Mercedes-Benz is constantly updating its engine control units to the state of the art and therefore reserves the right to introduce changes in design, equipment and technical features. You cannot, therefore, base any claims on the data, illustrations or descriptions in these Operating Instructions.

Mercedes-Benz AG

A Daimler AG company.

Document information	2
Applicability	2
List of versions	2
Revision levels	2
Introduction	3
Thank you for choosing a Mercedes-Benz product.	3
Table of contents	4
Safety	11
Key to symbols	11
General	11
Proper use	11
Requirements of personnel	11
Converting and modifying the MR2 control unit	12
Installation	12
Organizational actions	12
Safety precautions for engines with electronic control units	12
Original Mercedes-Benz parts	14
Safety and limp home program	14
Fault codes	14
List of abbreviations	15
List of abbreviations	16
Brief description	19
The MR2 diesel engine control unit	19
Control unit operating principle	19
Block diagram of control unit	20
Connecting the MR2 control unit to the vehicle	21
Configuration	22
Concept flexibility	22
Safeguards/redundancy:	22
Description of the inputs:	23
Description of the outputs:	23
Relationship between MR2 and FRE	23
Communication	23
MR2 <=> FRE coupled functions	24
Idle speed/engine speed/maximum speed control	24
Shutting off or throttling the engine via the FRE	24
Engine braking:	24
Engine stop, zero quantity:	24
Starter actuation starting lock, zero quantity:	24
Engine start and stop	25
Starter actuation (conditions)	25
Starter protection	25
Driver start	26
CAN start	26
Starter reset jumper	26
Starter output stage	26
CAN start by the vehicle control electronics (FRE)	27
Starting procedure	27
Service start button on engine block	27
Service stop button on the engine block	27
Cranking the engine via the service start and stop buttons	27
Revving the engine via the service start button	28
Stopping the engine	28
Terminal 50 plausibility check	28
Angle of delivery/start of delivery calculation	29

EOL detail correction	29
Controls (PID controller)	29
Operating modes	30
Working speed control	30
Controlled operation (normal operation)	30
Immobilizer	30
Tow-starting	30
Emergency operation	30
General overview of the Telligent engine system for the 500 model series	31
General overview of the Telligent engine system for the 900 model series	32
Description of OBD system	33
Drive cycle	33
Limp home mode	33
SCR exhaust gas aftertreatment system	33
Principle of exhaust gas aftertreatment	33
Method	34
SCRT exhaust gas aftertreatment system	35
Principle of exhaust gas aftertreatment	35
Function of the oxidation catalyst (DOC)	35
Function of the diesel particulate filter (DPF)	35
Function of the denox system (SCR catalyst)	35
Notes	36
Description of hardware	37
Description of mechanical components	37
Description of the control units for engine electronics	37
MR2A and MR2B control unit	37
Fuel cooling system	38
Example of model plate with barcode	40
MR2 version allocation table	41
Ambient conditions for mechanical components	41
General test conditions	41
Description of electrical components	42
Overview of system interfaces	42
55-pin engine connector for control unit MR2 A and B	43
16-pin vehicle connector for control unit MR2A and MR2B	46
Control unit power supply	47
Sensor system	48
Internal sensors	48
External sensors	48
Recording and evaluating the crankshaft/camshaft	55
Crankshaft/camshaft reference system	55
Cylinder actuation sequence	56
Cylinder order	56
Crank angle and TDC	57
Crankshaft and camshaft evaluation circuit	57
Synchronization of solenoid valve actuation	59
Digital inputs	60
Terminal 15	60
Terminal 50	61
Start/stop service switch	62
Heater flange	63
Actuator system	64
Unit pump (solenoid valve) characteristics	64
Unit pump actuation	64
Principle of unit pump actuation	64
The four actuation phases	64

Structure of the solenoid valve banks	66
Proportioning valve actuation	68
Communications interfaces	74
Diagnostics interface (K-line)	76
MR2 ground concept	78
Safety concept	78
Single-wire capability of engine CAN	78
System limp home capability	78
Engine protection	79
Control unit function monitoring (diagnosis)	79
Sensor monitoring	79
Monitoring the actuators	80
Fault memory	81
Description of software	82
Detection of hardware	82
Detection of MR2A and MR2B hardware version	82
Torque control	82
Structure/calculation of set torque	82
Map switching	84
Map switching process	84
Map switching: test function	85
Map switching: weighting function	85
Detection of engine state	85
"Stop" engine state	85
"Start" engine state	86
"Start abort" engine state	87
"On - idling" engine state	88
"On - controlled" engine state	88
"On - working speed control" engine state	89
"On - limp home" engine state	90
Requested torque gradient restriction	90
Limit torque calculation	90
Notes	93
Diagnosis	94
Scope of control unit diagnostic functions	94
Reading measurements	96
Table of measurements for MR2A hardware	96
Table of measurements for MR2B hardware	98
Reading binary values	101
Table of binary values for MR2A	101
Table of binary values for MR2B	104
Table of parameters	110
Notes on parameterization	119
Monitoring of parameter settings	120
General	120
MR parameterization for typical vehicle configurations:	120
Fault code tables	121
Assigning fault paths and fault types to new faults:	121
Fault paths	122
Fault types	125
Overview of fault MU IDs – MR2B fault memory	128
Overview of fault MU IDs – MR2A and MR2B fault memory	138
Overview of fault conditions	160
Overview of monitoring conditions	169
Overview of parameter configurations	174
Overview of scan cycles	175

Overview of debounce time	175
Causes of faults/pin assignments	176
MR2A: 55-pin multiple plug (engine connector)	176
MR2A: 16-pin multiple plug (vehicle connector)	185
MR2B: 55-pin multiple plug (engine connector)	187
MR2B: 16-pin multiple plug (vehicle connector)	198
OBD system	199
Monitored components/systems and fault codes	199
Storing the fault codes	199
Cumulative fault memory	199
Freeze frame data	199
Readiness status	200
Tester communication	200
Malfunction indicator (MI)	200
OBD component monitoring	201
Notes	203